ON A NOETHERIAN SPACE, M-PSEUDO-COHERENCE IS COHERENCE OF HIGHER COHOMOLOGY SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces to the representable case by etale descent of both sides, then transfers m-pseudo-coherence, coherence and vanishing along epsilon^* and concludes by the scheme-case lemma of Derived Categories of Schemes

Editorial note. The label lemma-identify-pseudo-coherent-noetherian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- quasi-compact algebraic spaces admit surjective etale maps from affine schemes (Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover)
- m-pseudo-coherence is insensitive to replacing the space by an etale cover (Cohomology on Sites, Lemma sites-cohomology-lemma-pseudo-coherent-independent-representative)
- coherence of sheaves is etale local on Noetherian spaces (Cohomology of Spaces, Lemma spaces-cohomology-lemma-coherent-Noetherian)
- the comparison of D_QCoh on the small etale site with the Zariski site of a scheme via epsilon, with E = epsilon^*E_0 (lemma-derived-quasi-coherent-small-etale-site of the same chapter)
- descent of m-pseudo-coherence, of finite type, and the flatness of epsilon (lemma-descend-pseudo-coherent, lemma-descend-finite-type and lemma-epsilon-flat of the same chapter)
- the scheme case: on a Noetherian scheme, m-pseudo-coherence is equivalent to coherence of H^i for i >= m and vanishing for i » 0 (Derived Categories of Schemes, Lemma perfect-lemma-identify-pseudo-coherent-noetherian)

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at proof line(s) 3356; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 3352; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08HD: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3081–3081. Prerequisite pointer invoked at proof line(s) 3360; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08HE: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3133–3133. Prerequisite pointer invoked at proof line(s) 3358; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08H8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 330–330. Prerequisite pointer invoked at proof line(s) 3362; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08E8: sources/stacks-chow/tags.txt, source lines 10664–10664. Prerequisite pointer invoked at proof line(s) 3365; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08FU: sources/stacks-chow/tags.txt, source lines 10722–10722. Prerequisite pointer invoked at proof line(s) 3344; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07UB: sources/stacks-chow/tags.txt, source lines 9966–9966. Prerequisite pointer invoked at proof line(s) 3347; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Prerequisite pointer invoked at proof line(s) 3339; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. For $m \in \mathbf{Z}$ the following are equivalent*

(1) *$H^i(E)$ is coherent for $i \geq m$ and zero for $i \gg 0$, and*
(2) *E is m-pseudo-coherent.*

In particular, E is pseudo-coherent if and only if E is an object of $D^-_{Coh}(\mathcal{O}_X)$.

Proof. As X is quasi-compact we can find an affine scheme U and a surjective étale morphism $U \to X$ (Properties of Spaces, Lemma 03H6). Observe that U is Noetherian. Note that E is m-pseudo-coherent if and only if $E|_U$ is m-pseudo-coherent (follows from the definition or from Cohomology on Sites, Lemma 08FU). Similarly, $H^i(E)$ is coherent if and only if $H^i(E)|_U = H^i(E|_U)$ is coherent (see Cohomology of Spaces, Lemma 07UB). Thus we may assume that X is representable.

If X is representable by a scheme X_0 then (Lemma 071Q) we can write $E = \epsilon^* E_0$ where E_0 is an object of $D_{QCoh}(\mathcal{O}_{X_0})$ and $\epsilon : X_{étale} \to (X_0)_{Zar}$ is as in (08H7). In this case E is m-pseudo-coherent if and only if E_0 is by Lemma 08HE. Similarly, $H^i(E_0)$ is of finite type (i.e., coherent) if and only if $H^i(E)$ is by Lemma 08HD.

Finally, $H^i(E_0) = 0$ if and only if $H^i(E) = 0$ by Lemma 08H8. Thus we reduce to the case of schemes which is Derived Categories of Schemes, Lemma 08E8. □